



ENDOCRINOLOGY

Adrenal Gland - 1

Internal Medicine Lecture Series

Objectives:

- ◆ Discussing the major manifestation of adrenal gland
- ◆ Increasing the level of knowledge about steroids use and side effects

Presenting problems in adrenal disease:

Cushing's syndrome:

- ◆ Cushing's syndrome is caused by excessive activation of glucocorticoid receptors.
- ◆ It is most commonly iatrogenic, due to prolonged administration of synthetic glucocorticoids such as prednisolone.
- ◆ Endogenous Cushing's syndrome is uncommon but is due to chronic over-production of cortisol by the adrenal glands, either as the result of an adrenal tumor or because of excessive production of ACTH by a pituitary tumor or ectopic ACTH production by other tumors.

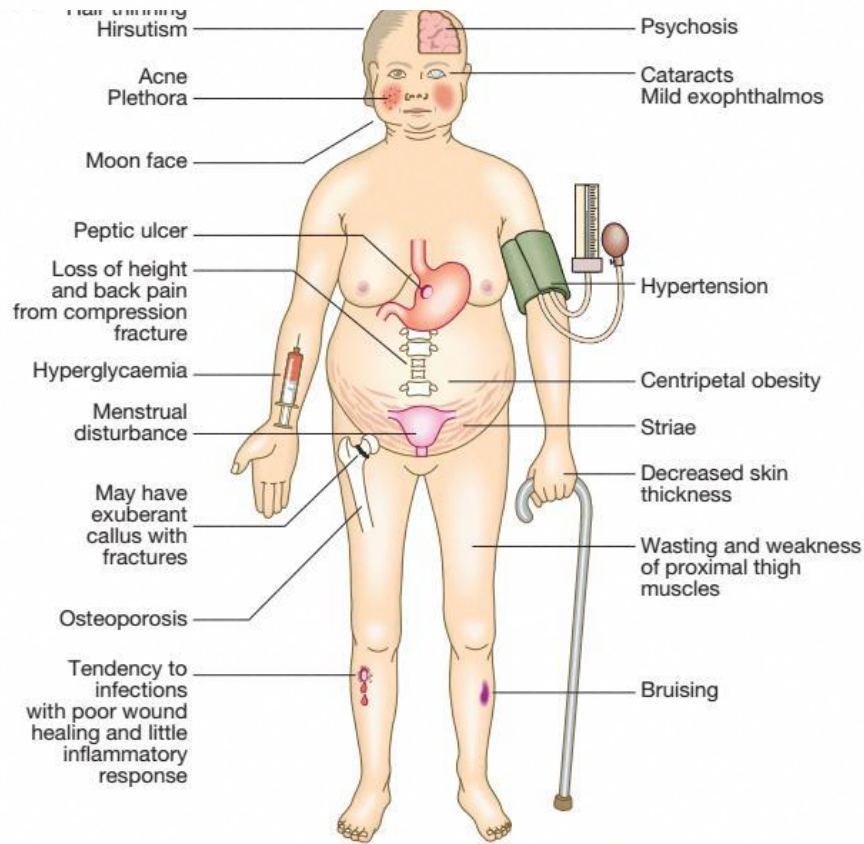
Aetiology:

- ◆ Amongst endogenous causes, pituitary-dependent cortisol excess (by convention, called Cushing's disease) accounts for approximately 80% of cases.

i	20.37 Classification of endogenous Cushing's syndrome
ACTH-dependent – 80%	
• Pituitary adenoma secreting ACTH (Cushing's disease) – 70% • Ectopic ACTH syndrome (bronchial carcinoid, small-cell lung carcinoma, other neuro-endocrine tumour) – 10%	
Non-ACTH-dependent – 20%	
• Adrenal adenoma – 15% • Adrenal carcinoma – 5% • ACTH-independent macronodular hyperplasia; primary pigmented nodular adrenal disease; McCune–Albright syndrome (together < 1%)	
Hypercortisolism due to other causes (also referred to as pseudo-Cushing's syndrome)	
• Alcohol excess (biochemical and clinical features) • Major depressive illness (biochemical features only, some clinical overlap) • Primary obesity (mild biochemical features, some clinical overlap)	
(ACTH = adrenocorticotrophic hormone)	

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Clinical assessment:

- ◆ Many of these are not specific to Cushing's syndrome and, because spontaneous Cushing's syndrome is rare, the positive predictive value of any single clinical feature alone is low.
- ◆ Moreover, some common disorders can be confused with Cushing's syndrome because they are associated with alterations in cortisol secretion: for example, obesity and depression.
- ◆ Features which favor Cushing's syndrome in an obese patient are bruising, myopathy and thin skin.
- ◆ It is vital to exclude iatrogenic causes in all patients with Cushing's syndrome since even inhaled or topical glucocorticoids can induce the syndrome in susceptible individuals.
- ◆ A careful drug history must therefore be taken before embarking on complex investigations.
- ◆ The high ACTH levels are associated with marked pigmentation because of binding to melanocortin 1 receptors on melanocytes in the skin.

kidney ,causing hypokalaemic alkalosis which aggravates myopathy and hyperglycemia (by inhibiting insulin secretion).

- ◆ When the tumour that is secreting ACTH is malignant, then the onset is usually rapid and may be associated with cachexia. For these reasons, the classical features of Cushing's syndrome are less common in ectopic ACTH syndrome; if present, they suggest that a less aggressive tumour, such as a bronchial carcinoid, is responsible.
- ◆ In Cushing's disease, the pituitary tumour is usually a microadenoma (< 10 mm in diameter); hence other features of a pituitary macroadenoma (hypopituitarism, visual failure or disconnection hyperprolactinaemia, are rare.

Investigations :

- ◆ Testing for Cushing's syndrome should be avoided under conditions of stress, such as an acute illness, because this activates the HPA axis, causing potentially spurious results.

The diagnosis of Cushing's is a two step process:

1. to establish whether the patient has Cushing's syndrome.
2. to define its cause .

Some additional tests are useful in all cases of Cushing's syndrome, including plasma electrolytes, glucose, glycosylated haemoglobin and bone mineral density measurement.

Establishing the presence of Cushing's syndrome :

Cushing's syndrome is confirmed by using two of three main tests:

1. failure to suppress serum cortisol with low doses of oral dexamethasone
2. loss of the normal circadian rhythm of cortisol, with inappropriately elevated late-night serum or salivary cortisol
3. increased 24-hour urine free cortisol

Dexamethasone is used for suppression testing because it does not cross- react in radioimmunoassays for cortisol.

dexamethasone at 2300 hrs and measuring serum cortisol at 0900 hrs the following day.

In a low-dose dexamethasone suppression test (LDDST), serum cortisol is measured following administration of 0.5 mg dexamethasone 4 times daily for 48 hours.

Cyclicity of cortisol secretion is a feature of all types of Cushing's syndrome and, if very variable, can confuse diagnosis.

Use of multiple salivary cortisol samples over weeks or months can be of use in diagnosis, but an elevated salivary cortisol alone should not be taken as proof of diagnosis.

In iatrogenic Cushing's syndrome, cortisol levels are low unless the patient is taking a corticosteroid (such as prednisolone) that cross-reacts in immunoassays with cortisol.

Determining the underlying cause:

- ◆ Once the presence of Cushing's syndrome is confirmed, measurement of plasma ACTH is the key to establishing the differential diagnosis.
- ◆ In the presence of excess cortisol secretion, an undetectable ACTH (below 1.1 pmol/L (5 pg/mL)) indicates an adrenal cause, while ACTH levels greater than 3.3 pmol/L (15 pg/mL) suggest a pituitary cause or ectopic ACTH.
- ◆ MRI detects around 60% of pituitary microadenoma secreting ACTH. If available, bilateral inferior petrosal sinus sampling (BIPSS) with measurement of ACTH is the best means of confirming Cushing's disease, unless MRI shows a tumor bigger than 6 mm. CT or MRI detects most adrenal tumors; adrenal carcinomas are usually large (> 5 cm) and have other features of malignancy .

- ◆ Untreated severe Cushing's syndrome has a 50% 5-year mortality.
- ◆ Most patients are treated surgically, but medical therapy may be given in severe cases for a few weeks prior to operation to improve the clinical state.
- ◆ A number of drugs are used to inhibit corticosteroid biosynthesis, including metyrapone and ketoconazole.
- ◆ The dose of these agents is best titrated against serum cortisol levels or 24-hour urine free cortisol

Cushing's disease:

- ◆ Trans-sphenoidal surgery carried out by an experienced surgeon with selective removal of the adenoma is the treatment of choice, with approximately 70% of patients going into immediate remission.
- ◆ Around 20% of patients suffer a recurrence, often years later, emphasising the need for life-long follow-up.
- ◆ Laparoscopic bilateral adrenalectomy performed by an expert surgeon effectively cures ACTH-dependent Cushing's syndrome, but in patients with pituitary dependent Cushing's syndrome, this can result in Nelson's syndrome. Laparoscopic bilateral adrenalectomy performed by an expert surgeon effectively cures ACTH-dependent Cushing's syndrome.
- ◆ The risk of Nelson's syndrome may be reduced by pituitary irradiation.

Adrenal tumours:

- ◆ Laparoscopic adrenal surgery is the treatment of choice for adrenal adenomas.
- ◆ Surgery offers the only prospect of cure for adrenocortical carcinomas, but in general, prognosis is poor with high rates of recurrence, even in patients with localized disease at presentation.
- ◆ Radiotherapy to the tumor bed reduces the risk of local recurrence, and systemic therapy consists of the adrenolytic drug mitotane and chemotherapy, but responses are often poor.

- ◆ Localized tumors, such as bronchial carcinoid, should be removed surgically.
- ◆ In patients with incurable malignancy, it is important to reduce the severity of the Cushing's syndrome using medical therapy or, if appropriate, bilateral adrenalectomy.

Therapeutic use of glucocorticoids:

- ◆ The remarkable anti-inflammatory properties of glucocorticoids have led to their use in a wide variety of clinical conditions but the hazards are significant.
- ◆ Topical preparations (dermal, rectal and inhaled) can also be absorbed into the systemic circulation, and although this rarely occurs to a sufficient degree to produce clinical features of Cushing's syndrome, it can result in significant suppression of endogenous ACTH and cortisol secretion.
- ◆ Severe Cushing's syndrome can result if there is concomitant administration of inhaled glucocorticoids and inhibitors of the liver enzyme CYP450 3A4, such as the antiretroviral drug ritonavir.



20.38 Approximate equivalent doses of glucocorticoids

- Hydrocortisone: 20 mg
- Cortisone acetate: 25 mg
- Prednisolone: 5 mg
- Dexamethasone: 0.5 mg

Adverse effects of glucocorticoids:

- ◆ Adverse effects are related to dose, duration of therapy, and pre-existing conditions that might be worsened by corticosteroid therapy, such as diabetes mellitus or osteoporosis.
- ◆ Osteoporosis is a particularly important problem because, for a given bone mineral density, the fracture risk is greater in glucocorticoid-treated patients than in post-menopausal osteoporosis.
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20.39 Advice to patients on glucocorticoid replacement therapy

Intercurrent stress

- Febrile illness: double dose of hydrocortisone

Surgery

- Minor operation: hydrocortisone 100 mg IM with pre-medication
- Major operation: hydrocortisone 100 mg 4 times daily for 24 hrs, then 50 mg IM 4 times daily until ready to take tablets

Vomiting

- Patients must have parenteral hydrocortisone if unable to take it by mouth

Steroid card

- Patient should carry this at all times; it should give information regarding diagnosis, steroid, dose and doctor

Bracelet and emergency pack

- Patients should be encouraged to buy a bracelet and have it engraved with the diagnosis, current treatment and a reference number for a central database
- Patients should be given a hydrocortisone emergency pack and trained in the self-administration of hydrocortisone 100 mg IM; they should be advised to take the pack on holidays/trips abroad

(IM = intramuscular)